

MATH2621 — Higher Complex Analysis. XXI

Laurent series

This lecture?

In this lecture, we prove Laurent's theorem about representing holomorphic functions in an annulus by a series, now called a Laurent series, and we give some examples of Laurent series.

[P]

An annulus is a set of the form $\{z \in \mathbb{C} : R_1 < |z - z_0| < R_2\}$; we allow R_1 to be 0 or R_2 to be ∞ .

A punctured ball is a set of the form $\{z \in \mathbb{C} : 0 < |z - z_0| < R\}$, and denoted by $B^\circ(z_0, R)$.

Laurent's theorem

Theorem

Suppose that A is the annulus $\{z \in \mathbb{C} : R_1 < |z - z_0| < R_2\}$ and $R_1 < r < R_2$. If $f \in H(A)$, then

$$f(w) = \sum_{n=-\infty}^{\infty} c_n (w - z_0)^n \quad \forall w \in A,$$

where

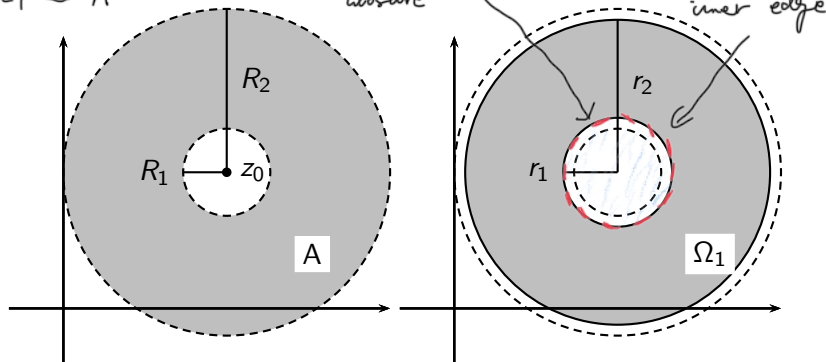
$$c_n = \frac{1}{2\pi i} \int_{\partial B(z_0, r)} \frac{f(z)}{(z - z_0)^{n+1}} dz.$$



Proof of Laurent's theorem

Proof. First, take r_1 and r_2 such that $R_1 < r_1 < r_2 < R_2$, and let Ω_1 be the domain $B(z_0, r_2) \setminus \overline{B}(z_0, r_1)$, as shown.

$\Omega_1 \subset A$



Suppose that $g \in H(A)$. Then $\int_{\partial\Omega_1} g(z) dz = 0$, by the Cauchy-Goursat theorem.

Proof of Laurent's theorem. 2

It follows that

$$\int_{\partial B(z_0, r_1)} g(z) dz = \int_{\partial B(z_0, r_2)} g(z) dz.$$

In particular, by taking $g(z) = f(z)/(z - z_0)^n$, where $f \in H(A)$, we see that the expression defining c_n does not depend on r .

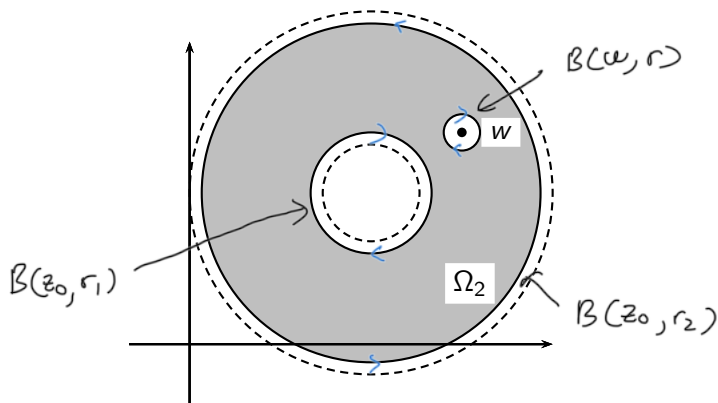
[P]

Now suppose that $f \in H(A)$ and $w \in A$. Choose r_1 and r_2 such that

$$R_1 < r_1 < |w - z_0| < r_2 < R_2,$$

and take a small ball $B(w, r)$ whose closure lies inside $B(z_0, r_2) \setminus \overline{B}(z_0, r_1)$, as shown below. Let $\Omega_2 = \Omega_1 \setminus \overline{B}(w, r)$.

Proof of Laurent's theorem. 3



[P] By the Cauchy–Goursat theorem,

$$\frac{1}{2\pi i} \int_{\partial\Omega_2} \frac{f(z)}{z - w} dz = 0.$$

Proof of Laurent's theorem. 4

Breaking the integral over the boundary of Ω_2 up into its three constituent integrals, we see that

$$\begin{aligned} & \frac{1}{2\pi i} \int_{\partial B(w,r)} \frac{f(z)}{z - w} dz \\ &= \frac{1}{2\pi i} \int_{\partial B(z_0, r_2)} \frac{f(z)}{z - w} dz - \frac{1}{2\pi i} \int_{\partial B(z_0, r_1)} \frac{f(z)}{z - w} dz. \end{aligned}$$

[P] From the Cauchy integral formula,

$$f(w) = \underbrace{\frac{1}{2\pi i} \int_{\partial B(z_0, r_2)} \frac{f(z)}{z - w} dz}_{\text{positive power}} - \underbrace{\frac{1}{2\pi i} \int_{\partial B(z_0, r_1)} \frac{f(z)}{z - w} dz}_{\text{negative power}}.$$

Proof of Laurent's theorem. 5

If z lies on $\partial B(z_0, r_2)$, then $|z - z_0| = r_2$ and $|w - z_0| < r_2$, so

$$\begin{aligned} & \int_{\partial B(z_0, r_2)} \frac{f(z)}{z - w} dz && \left| \frac{w - z_0}{z - z_0} \right| < 1 \quad \Leftrightarrow \\ &= \int_{\partial B(z_0, r_2)} \frac{f(z)}{(z - z_0) - (w - z_0)} dz && \left. \begin{array}{l} \uparrow \\ |w - z_0| < |z - z_0| \end{array} \right\} \\ &= \int_{\partial B(z_0, r_2)} \frac{f(z)}{(z - z_0)(1 - (w - z_0)/(z - z_0))} dz \\ &= \int_{\partial B(z_0, r_2)} \frac{f(z)}{(z - z_0)} \sum_{n=0}^{\infty} \frac{(w - z_0)^n}{(z - z_0)^n} dz \\ &= \sum_{n=0}^{\infty} (w - z_0)^n \int_{\partial B(z_0, r_2)} \frac{f(z)}{(z - z_0)^{n+1}} dz. \end{aligned}$$

Proof of Laurent's theorem. 6

$$\begin{aligned} - \int_{\partial B(z_0, r_1)} \frac{f(z)}{z - w} dz &= \int_{\partial B(z_0, r_1)} \frac{f(z)}{(w - z_0) - (z - z_0)} dz \\ &= \int_{\partial B(z_0, r_1)} \frac{f(z)}{(w - z_0)(1 - (z - z_0)/(w - z_0))} dz \\ &= \int_{\partial B(z_0, r_1)} \frac{f(z)}{(w - z_0)} \sum_{m=0}^{\infty} \frac{(z - z_0)^m}{(w - z_0)^m} dz \\ &= \sum_{m=0}^{\infty} \frac{1}{(w - z_0)^{m+1}} \int_{\partial B(z_0, r_1)} f(z) (z - z_0)^m dz \\ &= \sum_{n=-\infty}^{-1} (w - z_0)^n \int_{\partial B(z_0, r_1)} \frac{f(z)}{(z - z_0)^{n+1}} dz \end{aligned}$$

(at the last step, we substituted $n = -m - 1$).

Proof of Laurent's theorem. 7

It follows that

$$f(w) = \sum_{n=-\infty}^{\infty} c_n (w - z_0)^n \quad \forall w \in A,$$

where

$$c_n = \begin{cases} \frac{1}{2\pi i} \int_{\partial B(z_0, r_2)} \frac{f(z)}{(z - z_0)^{n+1}} dz & \text{if } n \geq 0 \\ \frac{1}{2\pi i} \int_{\partial B(z_0, r_1)} \frac{f(z)}{(z - z_0)^{n+1}} dz & \text{if } n \leq -1. \end{cases}$$

To conclude, we recall that the integrals defining c_n around $\partial B(z_0, r)$ do not depend on r in the range (R_1, R_2) . □

Finding Laurent series

A series in powers of $(z - z_0)$ that converges in an annulus with centre z_0 is called a *Laurent series*. We do not always find Laurent series by computing integrals; rather, often we compute integrals by finding Laurent series.

Exercise 1

Suppose that

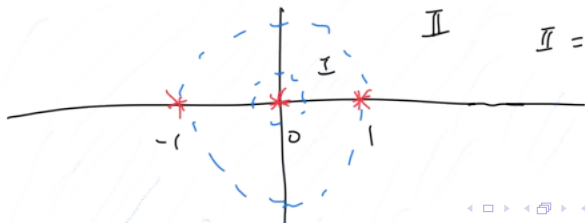
$$f(z) = \frac{1}{z(z^2 - 1)} = \frac{1}{z(z-1)(z+1)} = \frac{-1}{z} + \frac{1/2}{z-1} + \frac{1/2}{z+1}.$$

Find Laurent series for f in the largest annuli with centre 0 in which f is holomorphic.

Answer. [P] The function f has singularities at $0, \pm 1$. The largest annuli with centre 0 in which f is holomorphic are

$\{z \in \mathbb{C} : 0 < |z| < 1\}$ and $\{z \in \mathbb{C} : 1 < |z|\}$. $I = \{z : 0 < |z| < 1\}$

$$II = \{z : 1 < |z| < \infty\}$$



Answer to Exercise 1

In the annulus $\{z \in \mathbb{C} : 0 < |z| < 1\}$, also known as the punctured ball $B^\circ(0, 1)$, we may write

$$|z| < 1$$

$$\begin{aligned} f(z) &= \frac{-1}{z} - \frac{1/2}{1-z} + \frac{1/2}{1+z} \quad \downarrow \text{geometric series} \\ &= \frac{-1}{z} - \frac{1}{2} \left(1 + z + z^2 + z^3 + z^4 + \dots \right) \\ &\quad + \frac{1}{2} \left(1 - z + z^2 - z^3 + z^4 - \dots \right) \\ &= \frac{-1}{z} - \left(z + z^3 + z^5 + \dots \right) \\ &= (-1) \left(z^{-1} + z + z^3 + z^5 + \dots \right). \end{aligned}$$

Answer to Exercise 1 (continued)

In the annulus $\{z \in \mathbb{C} : 1 < |z| < \infty\}$, we may write

$$\begin{aligned} f(z) &= \frac{-1}{z} + \frac{1}{2z} \cdot \frac{1}{1-1/z} + \frac{1}{2z} \cdot \frac{1}{1+1/z} & \left| \frac{1}{z} \right| < 1 \\ &= \frac{-1}{z} + \frac{1}{2z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \frac{1}{z^4} + \dots \right) & \Leftrightarrow |z| > 1 \\ &\quad + \frac{1}{2z} \left(1 - \frac{1}{z} + \frac{1}{z^2} - \frac{1}{z^3} + \frac{1}{z^4} - \dots \right) \\ &= \frac{-1}{z} + \frac{1}{z} \left(1 + \frac{1}{z^2} + \frac{1}{z^4} + \frac{1}{z^6} + \dots \right) \\ &= \frac{1}{z} \left(\frac{1}{z^2} + \frac{1}{z^4} + \frac{1}{z^6} + \dots \right) \\ &= \left(\frac{1}{z^3} + \frac{1}{z^5} + \frac{1}{z^7} + \dots \right). \end{aligned}$$

△

Remarks

This form of presentation is legitimate as long as there are sufficiently many terms that the pattern is clear.

[P]

Note that both series have only odd powers, reflecting the symmetry of f ; indeed, z^n is an odd or even function as n is an odd or even integer. Note also that the radius of convergence of the first power series is 1; this corresponds to the fact that f is holomorphic in $B^\circ(0, 1)$ but not in any larger punctured ball with centre 0. The second series converges when $|z| > 1$; indeed, it is a geometric series with ratio $|1/z^2|$.

[P]

Note also that the first Laurent series has only one negative power of z , while the second has many, starting with z^{-3} . This is because $f(z)$ behaves like z^{-1} when z is small and like z^{-3} when z is large.

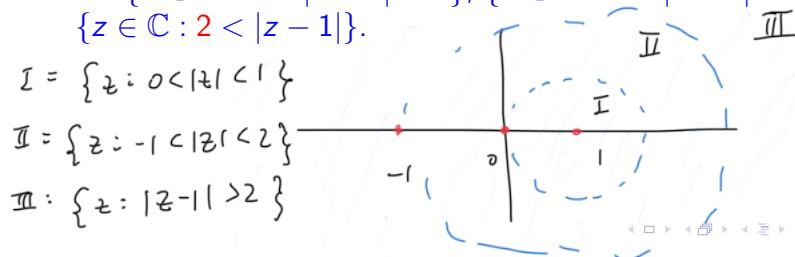
Exercise 2

Suppose that

$$f(z) = \frac{1}{z(z^2 - 1)} = \frac{1}{z(z-1)(z+1)} = \frac{-1}{z} + \frac{1/2}{z-1} + \frac{1/2}{z+1}.$$

Find Laurent series for f in the largest annuli with centre **1** in which f is holomorphic.

Answer. [P] The function f has singularities at $0, \pm 1$. The three largest annuli with centre 1 in which f is holomorphic are $\{z \in \mathbb{C} : 0 < |z-1| < 1\}$, $\{z \in \mathbb{C} : 1 < |z-1| < 2\}$, and $\{z \in \mathbb{C} : 2 < |z-1|\}$.



Answer to Exercise 2

In the annulus $\{z \in \mathbb{C} : 0 < |z - 1| < 1\}$, we may write

$$\begin{aligned}f(z) &= \frac{-1}{z} + \frac{1/2}{z-1} + \frac{1/2}{z+1} \\&= \frac{-1}{z-1+1} + \frac{1}{2} \cdot \frac{1}{z-1} + \frac{1}{2} \cdot \frac{1}{z-1+2} \\&= \frac{-1}{1+(z-1)} + \frac{1}{2} \cdot \frac{1}{z-1} + \frac{1}{4} \cdot \frac{1}{1+(z-1)/2} \\&= \sum_{n=0}^{\infty} (-1)^{n+1} (z-1)^n + \frac{1}{2} \cdot \frac{1}{z-1} + \frac{1}{4} \sum_{n=0}^{\infty} \frac{(-1)^n (z-1)^n}{2^n} \\&= \frac{1}{2} \cdot \frac{1}{z-1} + \sum_{n=0}^{\infty} (-1)^n \left(-1 + \frac{1}{2^{n+2}}\right) (z-1)^n \\&= \sum_{n=-1}^{\infty} (-1)^{n+1} \left(\frac{2^{n+2} - 1}{2^{n+2}}\right) (z-1)^n.\end{aligned}$$

Answer to Exercise 2 (continued)

In the annulus $\{z \in \mathbb{C} : 1 < |z - 1| < 2\}$, we may write

$$\begin{aligned}f(z) &= \frac{-1}{z} + \frac{1/2}{z-1} + \frac{1/2}{z+1} \\&= \frac{-1}{z-1+1} + \frac{1}{2} \cdot \frac{1}{z-1} + \frac{1}{2} \cdot \frac{1}{z-1+2} \\&= \frac{-1}{z-1} \cdot \frac{1}{1+1/(z-1)} + \frac{1}{2} \cdot \frac{1}{z-1} + \frac{1}{4} \cdot \frac{1}{1+(z-1)/2} \\&= \frac{-1}{z-1} \sum_{n=0}^{\infty} \frac{(-1)^n}{(z-1)^n} + \frac{1}{2} \cdot \frac{1}{z-1} + \frac{1}{4} \sum_{n=0}^{\infty} \frac{(-1)^n (z-1)^n}{2^n} \\&= \sum_{n=1}^{\infty} \frac{(-1)^n}{(z-1)^n} + \frac{1}{2} \cdot \frac{1}{z-1} + \sum_{n=0}^{\infty} \frac{(-1)^n (z-1)^n}{2^{n+2}} \\&= \sum_{n=-\infty}^{\infty} c_n (z-1)^n,\end{aligned}$$

where $c_n = (-1)^n$ if $n \leq -2$ and $c_n = (-1)^n 2^{-n-2}$ if $n \geq -1$.

Answer to Exercise 2 (continued)

In the annulus $\{z \in \mathbb{C} : 2 < |z - 1| < \infty\}$, we may write

$$\begin{aligned}f(z) &= \frac{-1}{z} + \frac{1/2}{z-1} + \frac{1/2}{z+1} \\&= \frac{-1}{z-1+1} + \frac{1}{2} \cdot \frac{1}{z-1} + \frac{1}{2} \cdot \frac{1}{z-1+2} \\&= \frac{-1}{z-1} \cdot \frac{1}{1+1/(z-1)} + \frac{1}{2} \cdot \frac{1}{z-1} + \frac{1}{2(z-1)} \cdot \frac{1}{1+2/(z-1)} \\&= \frac{-1}{z-1} \sum_{n=0}^{\infty} \frac{(-1)^n}{(z-1)^n} + \frac{1}{2} \cdot \frac{1}{z-1} + \frac{1}{2(z-1)} \sum_{n=0}^{\infty} \frac{(-2)^n}{(z-1)^n} \\&= \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{(z-1)^{n+1}} + \frac{1}{2} \cdot \frac{1}{z-1} + \sum_{n=0}^{\infty} \frac{(-1)^n 2^{n-1}}{(z-1)^{n+1}} \\&= \sum_{m=3}^{\infty} \frac{(-1)^{m-1} (2^{m-2} - 1)}{(z-1)^m}.\end{aligned}$$

△

Remarks

These series involve both odd and even powers, reflecting the fact that f is neither even nor odd about 1.

None of these series converges in any larger annulus, since otherwise f would be holomorphic in a larger domain.

Note also that the last Laurent series starts with z^{-3} . This corresponds to the fact that $f(z)$ behaves like z^{-3} at infinity, and that $(z - 1)^{-3}$ and z^{-3} behave similarly there. Thus we should have expected cancellations in the terms $(z - 1)^{-1}$ and $(z - 1)^{-2}$.